TELECOM CUSTOMER CHURN ANALYSIS USING MACHINE LEARNING

Introduction

Customer churn is a major challenge in the telecom industry, where customers frequently switch service providers due to pricing, service quality, or contract flexibility. Predicting churn helps telecom companies proactively retain customers and reduce revenue loss. This project focuses on analyzing customer data and building a predictive model to identify customers who are likely to churn.

Abstract

The objective of this project is to predict customer churn using machine learning techniques and derive actionable business insights. The dataset contains customer demographics, account information, contract details, and billing information. Data preprocessing, exploratory data analysis (EDA), feature encoding, and model training were performed using Python. A Random Forest classifier was implemented to predict churn. Model performance was evaluated using accuracy, precision, recall, and confusion matrix. SHAP analysis was used to interpret feature importance. The results provide strategic recommendations to reduce churn and improve customer retention.

Tools & Technologies Used

- Python
- Google Colab
- Pandas & NumPy (Data Processing)
- Matplotlib & Seaborn (Visualization)
- Scikit-learn (Machine Learning)
- SHAP (Model Explainability)
- SQL (Data Aggregation)
- PowerPoint (Presentation)
- GitHub (Version Control)

Steps Involved in Building the Project

Step 1: Data Collection

The Telco Customer Churn dataset was obtained from Kaggle. **Step 2:**

Data Cleaning

- Removed missing values
- Converted TotalCharges to numeric
- Encoded categorical variables using LabelEncoder **Step 3: Exploratory Data**

Analysis

- Analyzed churn rate by contract type
- Studied relationship between tenure and churn
- Compared monthly charges with churn probability

Step 4: Feature Engineering

- Converted categorical features into numerical format
- Separated independent and dependent variables **Step 5: Model Building**
- Split data into training and testing sets
- Implemented Random Forest Classifier
- Trained model on training dataset

Step 6: Model Evaluation

- Generated classification report
- Built confusion matrix
- Measured accuracy, precision, recall **Step 7: Model Explainability**
- Used SHAP to identify important features influencing churn
- Interpreted feature impact for business decisions

Results & Insights

- Model achieved approximately 80–85% accuracy
- Customers with month-to-month contracts show higher churn
- Higher monthly charges increase churn probability

- Customers with short tenure are more likely to leave

Business Recommendations

- Offer loyalty discounts for long-term contracts
- Introduce retention offers for high-risk customers
- Provide incentives for automatic payment methods
- Improve service quality for short-tenure customers

Conclusion

This project successfully predicted customer churn using machine learning techniques. The integration of SHAP provided explainable insights, helping bridge the gap between technical predictions and business decisions.